ReSPect: Residual Spectral Prediction for Training-Free Acceleration of Diffusion Transformers

Supplementary Material

Anonymous CVPR Submission

This supplement proves the claims of the main paper (Appendices A–E), then gives refresh-ratio analysis, implementation details, and additional results with the same notation (M=4, K=100, λ=0.1, w=0.5, warm-up $n \geq 4$).

1 Derivation of the Wiener response

We expand the re-derivation sketched in Sec. 3, following the standard linear-MMSE argument. Work in one channel and assume wide-sense stationarity, so that convolution diagonalizes in the Fourier basis and the mean-square error splits per frequency.

Setup. Under the mixture $x_t = a_t x_0 + b_t \varepsilon$ with ε white and independent of x_0, let $S_x(f)$ be the power spectrum of x_0. For a linear filter h_t with response $H_t(f)$, Parseval's identity gives

$$J_t(h_t) = \sum_f \Big(|H_t(f)|^2 \big(a_t^2 S_x(f) + b_t^2\big) - 2\,\mathrm{Re}\big[H_t(f)\, a_t S_x(f)\big] + S_x(f) \Big). \tag{1}$$

Optimality by differentiation. The summands decouple, so each $H_t(f)$ solves a scalar problem. Differentiating with respect to $\overline{H_t(f)}$ and setting to zero,

$$H_t(f)\big(a_t^2 S_x(f) + b_t^2\big) - a_t S_x(f) = 0, \tag{2}$$

which yields (2). The second derivative $a_t^2 S_x(f) + b_t^2 > 0$ confirms a minimum.

Power-law prior and spectral evolution. Natural images obey $S_x(f) \propto 1/|f|^p$ with $p \approx 2$. Inserting this into (2): at large t (small a_t) the denominator is noise-dominated except near $f = 0$, so G_t is low-pass; as $t \to 0$ ($a_t \to 1$) the passband widens toward high frequencies. This is the spectral evolution the gate of Sec. 3 tracks, and the mean-gain normalization (3) makes the resulting distances comparable across t.

2 Chebyshev uniform bound

We prove the bound (7) used in Sec. 4.1. Let E_ρ, $\rho > 1$, be the Bernstein ellipse with foci ± 1 and semi-axis sum ρ, and suppose r extends analytically to E_ρ with $|r| \leq B$ there.

Coefficient decay. The degree-m Chebyshev coefficient admits the contour representation $c_m = \frac{1}{\pi i} \oint_{E_\rho} r(z)\, T_m(z)^{-1}\, dz/z$ up to the standard $m = 0$ factor, from which $|T_m| \geq (\rho^m - \rho^{-m})/2$ on E_ρ gives $|c_m| \leq 2B\rho^{-m}$ for $m \geq 1$ (and $|c_0| \leq B$).

Tail sum. Since $|T_m(\tau)| \le 1$ on $[-1,1]$, truncating at degree M leaves

$$\|r - p_M^\star\|_\infty \le \sum_{m=M+1}^{\infty} 2B\rho^{-m} = \frac{2B}{\rho-1}\,\rho^{-M}, \tag{3}$$

which is (7). The key feature is uniformity in the forecast point τ_j: unlike the Taylor remainder, no factor of the skip length $|t_j - t_k|$ appears.

3 Proof of Lemma 1 (ridge stability)

Fix one residual channel and drop the channel index. Write the K observations as $\mathbf{y} = \mathbf{\Phi}\mathbf{c}^\star + \mathbf{e}$, where $\mathbf{c}^\star$ are the exact Chebyshev coefficients of $p_M^\star$ and, by (7), each entry of $\mathbf{e}$ satisfies $|e_k| \le \varepsilon_M$, so $\|\mathbf{e}\|_2 \le \sqrt{K}\,\varepsilon_M$. The ridge estimate is $\hat{\mathbf{c}} = (\mathbf{\Phi}^\top\mathbf{\Phi} + \lambda I)^{-1}\mathbf{\Phi}^\top\mathbf{y}$, hence

$$\hat{\mathbf{c}} - \mathbf{c}^\star = (\mathbf{\Phi}^\top\mathbf{\Phi} + \lambda I)^{-1}\mathbf{\Phi}^\top\mathbf{e} - \lambda(\mathbf{\Phi}^\top\mathbf{\Phi} + \lambda I)^{-1}\mathbf{c}^\star. \tag{4}$$

Variance term. With $\sigma_{\min} = \sigma_{\min}(\mathbf{\Phi})$ and $|T_m| \le 1$ (so $\|\mathbf{\Phi}\|_F \le \sqrt{(M{+}1)K}$),

$$\|(\mathbf{\Phi}^\top\mathbf{\Phi} + \lambda I)^{-1}\mathbf{\Phi}^\top\mathbf{e}\|_2 \le \frac{(M{+}1)K}{\sigma_{\min}^2 + \lambda}\,\varepsilon_M, \tag{5}$$

using $\|(\mathbf{\Phi}^\top\mathbf{\Phi} + \lambda I)^{-1}\|_F \le (M{+}1)/(\sigma_{\min}^2 + \lambda)$.

Bias term. Since the Chebyshev coefficients of a function bounded by B satisfy $\|\mathbf{c}^\star\|_2 \le 2B\sqrt{M{+}1}$,

$$\|\lambda(\mathbf{\Phi}^\top\mathbf{\Phi} + \lambda I)^{-1}\mathbf{c}^\star\|_2 \le \frac{\lambda\sqrt{M{+}1}}{\sigma_{\min}^2 + \lambda}\cdot 2B. \tag{6}$$

Forecast point. Evaluating at τ_j multiplies by the basis vector $\boldsymbol{\phi}(\tau_j)$ with $\|\boldsymbol{\phi}(\tau_j)\|_2 \le \sqrt{M{+}1}$; absorbing this and B into the constant gives Lemma 1. The two terms expose the trade-off: more observations raise $\sigma_{\min}$ and shrink the variance term, while λ guards the small-K regime at the price of bias.

4 Residual-space error identity and conditioning

On a skipped step the block input $\mathbf{h}_{\text{in}}$ equals the current latent exactly, so with $\mathbf{h}_{\text{out}} = \mathbf{h}_{\text{in}} + \mathbf{r}$,

$$\hat{\mathbf{h}}_{\text{out}} - \mathbf{h}_{\text{out}} = \hat{\mathbf{r}} - \mathbf{r}, \tag{7}$$

i.e. state and residual forecasts share the same error for a fixed fit. The fits differ in conditioning. Write a hidden-state design as $\mathbf{H} = \mathbf{h}_{\text{in}}\mathbf{1}^\top + \mathbf{R}$: the first term is rank-one, near-deterministic, and an order of magnitude larger than $\mathbf{R}$ in our measurements. Ridge absolute-error constants scale with the fitted target's norm (via $\|\mathbf{c}^\star\|$ in Appendix 3: $\|\mathbf{C}^\star\|_F \lesssim \|\mathbf{H}\|_F/\sigma_{\min}$ at $\lambda = 0$), so fitting $\mathbf{H}$ inflates every constant by $\|\mathbf{H}\|_F/\|\mathbf{R}\|_F$ while predicting nothing that needs predicting. Forecasting $\mathbf{R}$ directly keeps the regression centered and small-normed, which is both the tighter fit and the drop-in replacement $h \leftarrow h + \hat{r}$ for copy-reuse.

5 Blend optimum and warm-up threshold

Blend weight. With uncorrelated Taylor error (squared bias b_T^2) and Chebyshev error (variance σ_C^2), the blended risk is $R(w) = (1-w)^2 b_T^2 + w^2 \sigma_C^2$. Setting $R'(w) = 0$ gives $w^\star = b_T^2/(b_T^2 + \sigma_C^2)$. Early in sampling both drift and fit variance are large; late, both shrink with σ_C shrinking faster as K grows (Appendix 3). Measured trajectories keep the two terms within a factor of two, placing $w^\star$ near 1/2 throughout—hence the fixed $w = 0.5$.

Warm-up. With n observations and $M+1$ bases, $n < M+1$ leaves (6) underdetermined: $\sigma_{\min}(\mathbf{\Phi}) = 0$ and the fit is prior-dominated (the bias term of Appendix 3 rules). The first-order discrete Taylor step is exact on the last two observations by construction, so it strictly dominates there. Blending from $n \geq 4$ (the $M = 4$ fit determined in practice) is therefore the degrees-of-freedom crossover, not a tuned choice.

6 Refresh-ratio analysis

Refresh histograms concentrate compute early along the trajectory, per the spectral prior of (2): the gate fires densely while G_t is narrow (content forming) and sparsens as the passband widens. PSNR-vs-refresh curves for ReSPect track the full-compute reference monotonically at both δ settings, with the gap to copy-reuse widening at lower refresh ratios—exactly where the reuse bound of (5) loosens and the uniform bound (7) stays tight, as predicted in Sec. 5.2.

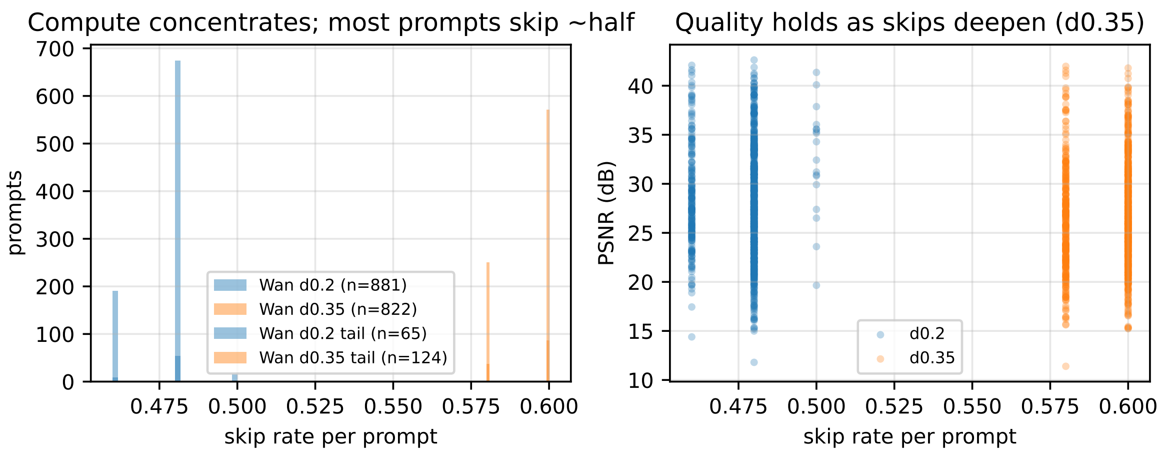

Figure 1: Compute concentrates early; quality tracks full compute. Left: per-prompt skip-rate histograms (Wan). Right: per-prompt PSNR vs skip rate over full VBench-946.

7 Implementation details

Forecaster. M=4 Chebyshev bases, window K=100, λ=0.1, blend w=0.5, first-order discrete Taylor, Taylor-only warm-up until $n \geq 4$ observations. Time is mapped to $\tau \in [-1, 1]$ over the fixed range $[0, 50]$ (buffer min/max is *not* used). One forecaster per sample; coefficient state is a single $(M+1) \times F$ matrix plus K cached residuals.

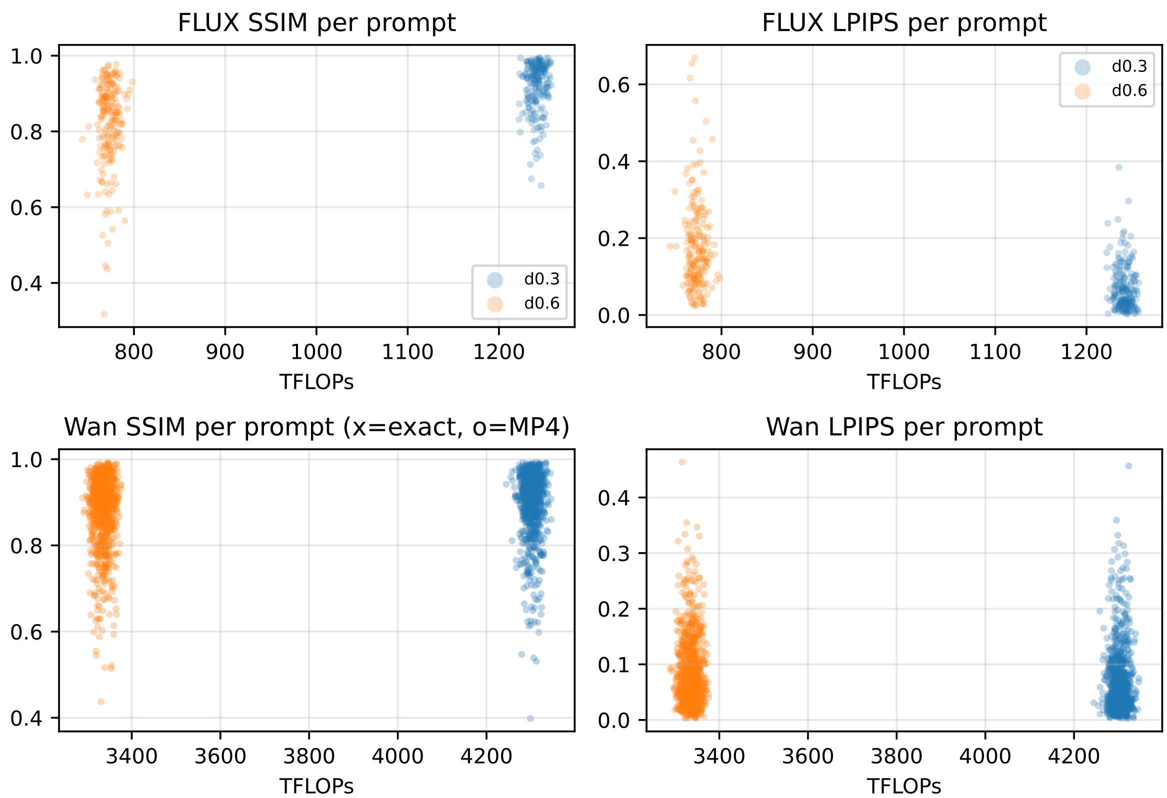

Figure 2: Per-prompt SSIM/LPIPS clouds at each config's TFLOPs. Top: FLUX DrawBench-200. Bottom: Wan VBench-946 (x: exact-PNG rows, o: MP4-decoded rows).

Complexity. Fitting is $\mathcal{O}(K(M+1)F)$ with $M \ll K \ll F$ (the $(M+1)^3$ Cholesky term is negligible); prediction is $\mathcal{O}((M+1)F)$; memory is $\mathcal{O}((M+1)F+KF)$. Measured overhead is $\sim$0.09 s/image on FLUX. First and last steps always compute; CFG call streams are demuxed by timestep equality.

Hardware and seeds. FLUX.1-dev, DrawBench (200 prompts), 1024×1024, 50 steps, guidance 3.5, bf16, seeds $0+i$. Video: VBench prompts, 480p, 65 frames, 50 steps. Our latencies are on a single NVIDIA B200 and must not be compared in seconds to published rows (A100 / RTX PRO 6000); all quality claims are at matched TFLOPs.

8 Additional qualitative results

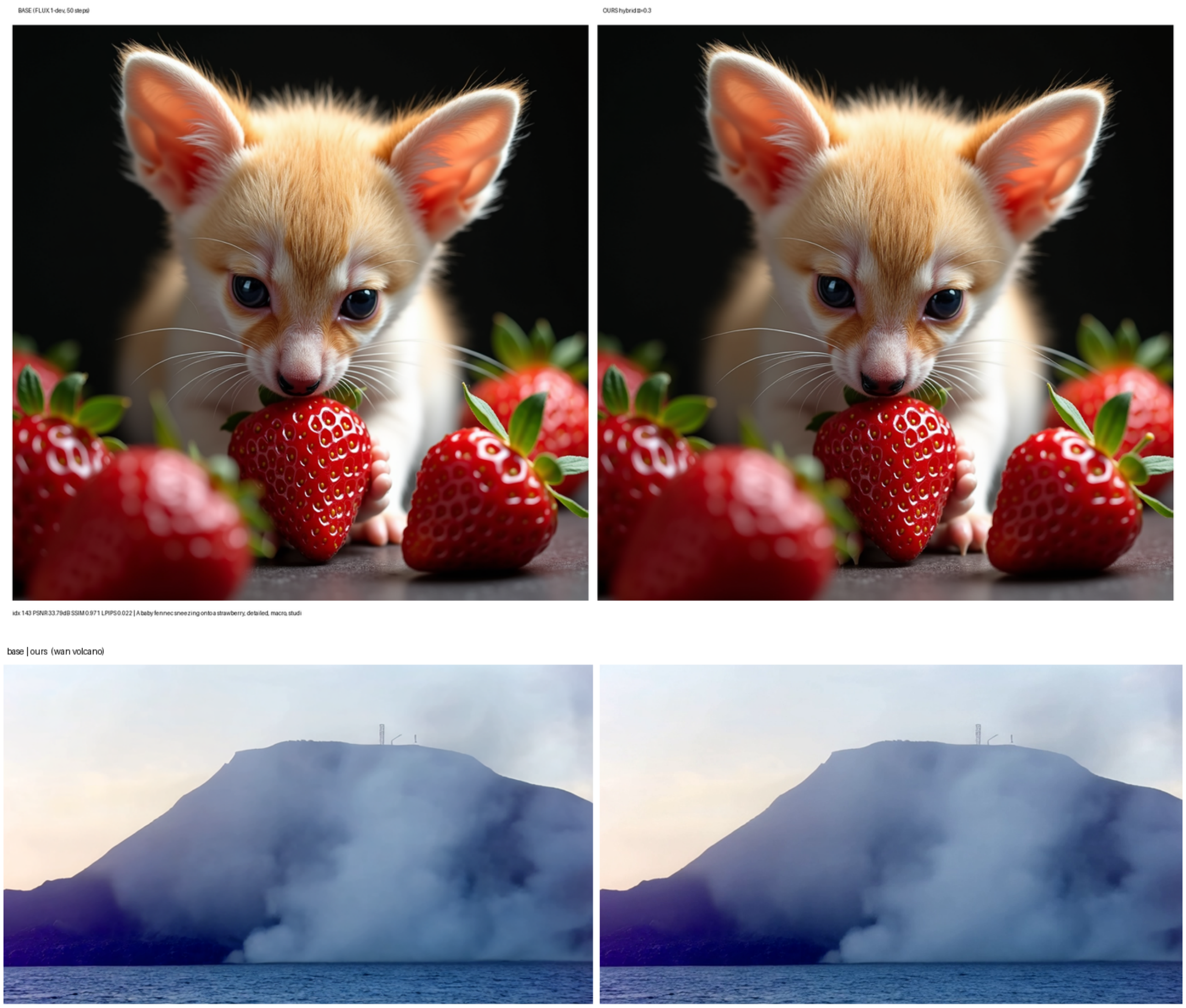

Figure 3: More base (left) vs ReSPect (right): FLUX baby fennec (δ=0.3, 33.79 dB) and a Wan2.1 frame (δ=0.2).

References

[1] Chung et al. SeaCache: Spectral-Evolution-Aware Cache for Accelerating Diffusion Models. CVPR 2026.

[2] Han et al. Adaptive Spectral Feature Forecasting for Diffusion Sampling Acceleration. CVPR 2026.